总结：

三角函数导数：

$\sin x = \cos x$

$\cos x = -\sin x$

$\tan x = \sec^2 x$

$\sec x = \sec x \tan x$

$\cot x = -\csc^2 x$

$\csc x = -\csc x \cot x$

$\sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$

$\cos^{-1} x = -\frac{1}{\sqrt{1-x^2}}$

$\tan^{-1} x = \frac{1}{1+x^2}$

$\cot^{-1} x = -\frac{1}{1+x^2}$

$\sec^{-1} x = \frac{1}{|x|\sqrt{x^2-1}}$

$\csc^{-1} x = -\frac{1}{|x|\sqrt{x^2-1}}$

不知道叫啥名

$\lim_{x\to 0} \frac{\sin x}{x} = 1$

$\lim_{x\to \infty} \frac{\sin x}{x} = 0$

$\lim_{x\to 0} \frac{1-\cos x}{x} = 0$

$\lim_{x\to 0} \frac{\tan x}{x} = 1$

恒等式：

$\sin^2 x + \cos^2 x = 1$

$1 + \tan^2 x = \sec^2 x$

$1 + \cot^2 x = \csc^2 x$

降幂公式：

$\sin^2 = \frac{1-\cos 2x}{2}$

$\cos^2 = \frac{1+\cos 2x}{2}$

二倍角：

$\sin 2x = 2\sin x \cos x$

反函数导数定理：

if $f^{-1}(a)=b$, $(f^{-1})'(a)=\frac{1}{f'(b)}$

步骤：① 求 b

② 求 $f'(x)$，代入 $x=b$ 得到 $f'(b)$

③ $(f^{-1})'(a)=\frac{1}{f'(b)}$

例题1：已知$f(x)=x^5+2x^3+x-1$，求$(f^{-1})'(3)$。

① 令$f(b)=3$，代入得$b^5+2b^3+b-1=3$，试算得$b=1$（$1+2+1-1=3$）

② 求$f'(x)=5x^4+6x^2+1$，则$f'(1)=5+6+1=12$；

③ $(f^{-1})'(3)=\frac{1}{12}$。

例二：表格给出$f(3)=-1$、$f'(3)=-4$，则$g'(-1)=$

$g'(-1)=\frac{1}{f'(3)}=-\frac{1}{4}$（其中$y=f^{-1}$）。

指数与对数的导数

指：$\frac{d}{dx}(e^x)=e^x$

$\frac{d}{dx}(e^u)=e^u\cdot\frac{du}{dx}$（u是x的可导函数）

一般指数函数：

① 化为自然指数：$a^x=e^{x\ln a}$

② 求导：$\frac{d}{dx}(a^x)=a^x\cdot\ln a$

复合：$\frac{d}{dx}(a^u)=a^u\cdot\ln a\cdot\frac{du}{dx}$

求$y=e^{x^2+3x}$的导数　令$u=x^2+3x$，则$\frac{du}{dx}=2x+3$，故$\frac{dy}{dx}=e^{x^2+3x}\cdot(2x+3)$

求$y=3^{\csc x}$的导数　令$u=\csc x$，则$\frac{du}{dx}=-\csc x\cot x$，故$\frac{dy}{dx}=3^{\csc x}\cdot\ln 3\cdot(-\csc x\cot x)$。

对：$\frac{d}{dx}(\ln x)=\frac{1}{x}$

$\frac{d}{dx}(\ln u)=\frac{1}{u}\cdot\frac{du}{dx}$

幂指函数求导：$y=u^v$，$u>0$，u、v均为x的可导函数

① 两边取 ln，$\ln y=v\ln u$

② 两边求导，$\frac{1}{y}\cdot\frac{dy}{dx}=\frac{dv}{dx}\cdot\ln u+v\cdot\frac{1}{u}\cdot\frac{du}{dx}$　$(\frac{1}{y}\cdot y'=v'\cdot\ln u+\frac{v}{u}\cdot u')$

③ $\frac{dy}{dx}=u^v\left(\frac{dv}{dx}\cdot\ln u+\frac{v}{u}\cdot\frac{du}{dx}\right)$　$(y'=u^v(v'\cdot\ln u+\frac{v}{u}\cdot u'))$

例题：求$y=x^x$（$x>0$）的导数。

① $\ln y=x\ln x$；② 求导得$\frac{1}{y}\cdot\frac{dy}{dx}=\ln x+x\cdot\frac{1}{x}=\ln x+1$；③ 故$\frac{dy}{dx}=x^x(\ln x+1)$。

求$y=\ln(\ln x)$的导数

令$u=\ln x$，则$\frac{du}{dx}=\frac{1}{x}$，故$\frac{dy}{dx}=\frac{1}{\ln x}\cdot\frac{1}{x}=\frac{1}{x\ln x}$

求$y=\log_3(1+x\ln 3)$的导数

令$u=1+x\ln 3$，则$\frac{du}{dx}=\ln 3$，故$\frac{dy}{dx}=\frac{1}{(1+x\ln 3)\ln 3}\cdot\ln 3=\frac{1}{1+x\ln 3}$

Related Rate

题目：水以$0.25\,m^3$/min的速率注入倒置圆锥水箱，水箱高$3m$，底面半径$1.5m$。当水深$1.8m$时，水位上升速率是多少？

步骤：

1. **定义变量**：设V为水箱内水的体积（m^3），h为水深（m），r为水面半径（m），t为时间（min）；水箱高$H=3m$，底面半径$R=1.5m$（常量）。
2. **已知条件**：$\frac{dV}{dt}=0.25\,m^3$/min，当$h=1.8m$时，求$\frac{dh}{dt}$。
3. **变量关系**：圆锥水箱的高与底面半径比为$\frac{H}{R}=\frac{3}{1.5}=2$，由相似三角形可知，水面半径与水深比相同，即$\frac{r}{h}=\frac{1}{2}$，故$r=\frac{h}{2}$；圆锥内水的体积$V=\frac{1}{3}\pi r^2 h$，代入$r=\frac{h}{2}$得$V=\frac{1}{3}\pi\left(\frac{h}{2}\right)^2 h=\frac{1}{12}\pi h^3$（消去$r$，仅含$V$与$h$）。
4. **对t求导**：$\frac{dV}{dt}=\frac{1}{12}\pi\cdot 3h^2\cdot\frac{dh}{dt}=\frac{1}{4}\pi h^2\cdot\frac{dh}{dt}$，整理得$\frac{dh}{dt}=\frac{4\cdot\frac{dV}{dt}}{\pi h^2}$。
5. **代入计算**：将$\frac{dV}{dt}=0.25$、$h=1.8$代入，得$\frac{dh}{dt}=\frac{4\times 0.25}{\pi\times(1.8)^2}=\frac{1}{\pi\times 3.24}\approx 0.098\,m$/min。

题目：测距仪距热气球起飞点$150m$，当测距仪仰角为$\frac{\pi}{4}$（45°）时，仰角以0.14 rad/min的速率增加，求此时热气球的上升速率。

步骤：

1. **定义变量**：设y为热气球高度（m），θ为测距仪仰角（rad），t为时间（min）；测距仪到起飞点距离$150m$（常量，水平距离，与y垂直）。
2. **已知条件**：当$\theta=\frac{\pi}{4}$时，$\frac{d\theta}{dt}=0.14$ rad/min，求$\frac{dy}{dt}$。
3. **变量关系**：由三角函数定义，$\tan\theta=\frac{y}{150}$，故$y=150\tan\theta$。
4. **对t求导**：$\frac{dy}{dt}=150\sec^2\theta\cdot\frac{d\theta}{dt}$（$\sec^2\theta=1+\tan^2\theta$）。
5. **代入计算**：当$\theta=\frac{\pi}{4}$时，$\sec^2\frac{\pi}{4}=2$，故$\frac{dy}{dt}=150\times 2\times 0.14=42\,m$/min。

点斜式：
$y-f(a)=f'(a)(x-a)$ $\Rightarrow$

Local Linearity & Linearization

$L(x)=f(a)+f'(a)(x-a)$

- 当 x 接近 a 时，$f(x)\approx L(x)$

应用及判断：

$\sqrt{4.2}$ ？：$f(x)=\sqrt{1+x}$ 在 $a=3$ 处的线性化

高估低估判断：

若 $f(x)$ 在 $x=a$ 附近凹向上（$f''(x)>0$），$L(x)$ 低估　　凹向下　　高估

① $f(3)=\sqrt{1+3}=2$

② $f'(x)=\frac{1}{2\sqrt{1+x}}$，$f'(3)=\frac{1}{4}$

③ $L(x)=2+\frac{1}{4}(x-3)$

④ $\sqrt{4.2}=\sqrt{1+3.2}$，$L(3.2)=2+\frac{1}{4}(3.2-3)=2.05$

⑤ $f''(x)=-\frac{1}{4}(1+x)^{-\frac{3}{2}}$，$x>-1$，$f''(x)<0$，concave down，高估

L'Hospitable Rule 判断(一转化)一应用一计算

① $\frac{0}{0}$ & $\frac{\infty}{\infty}$ 型

$f(a)=g(a)=0$ 或 $\pm\infty$，f 和 g 在含 a 的开区间内可导，$g'(x)\neq 0$，$x\neq a$：

$$\lim_{x\to a}\frac{f(x)}{g(x)}=\lim_{x\to a}\frac{f'(x)}{g'(x)}$$

如果求完还是不定式，一直求到不是再代入.

② $\infty\cdot 0$ 型

当 $x\to a$（或∞），$\lim f(x)=\infty$ 且 $\lim g(x)=0$，则 $\lim[f(x)\cdot g(x)]$ 为 $\infty\cdot 0$ 型

转化：

取倒数 $\infty\cdot 0=\frac{0}{\frac{1}{\infty}}=\frac{0}{0}$ 或 $\infty\cdot 0=\frac{\infty}{\frac{1}{0}}=\frac{\infty}{\infty}$

③ $\infty-\infty$ 型

$\lim_{x\to a(\text{或}\infty)} f(x)=\infty$ 且 $\lim_{x\to a(\text{或}\infty)} g(x)=\infty$，则 $\lim[f(x)-g(x)]$ 为 $\infty-\infty$

转化：通分/有理化 ← 根式

↑分式

例题：求 $\lim_{x\to+\infty}\left(\sqrt{x^2+2x}-x\right)$

步骤1：判断类型

当 $x\to+\infty$ 时，$\sqrt{x^2+2x}=\sqrt{x^2\left(1+\frac{2}{x}\right)}=x\sqrt{1+\frac{2}{x}}\to+\infty$，同时 $x\to+\infty$，因此 $\sqrt{x^2+2x}-x$ 是"无穷大减无穷大"，属于 $\infty-\infty$ 型，需有理化转化。

步骤2：有理化（消去根号）

对 $\sqrt{x^2+2x}-x$ 乘以其"共轭式" $\sqrt{x^2+2x}+x$（分子分母同乘，值不变），利用平方差公式展开：

$$\sqrt{x^2+2x}-x=\frac{(\sqrt{x^2+2x}-x)\cdot(\sqrt{x^2+2x}+x)}{\sqrt{x^2+2x}+x}$$

分子展开：$\left(\sqrt{x^2+2x}\right)^2-x^2=(x^2+2x)-x^2=2x$，因此原式化简为：

$$\lim_{x\to+\infty}\frac{2x}{\sqrt{x^2+2x}+x}$$

步骤3：转化为基础型（$\frac{\infty}{\infty}$型）

④、1^∞ 型

当 $x\to a$ 或 ∞，$\lim f(x)=1$，$\lim g(x)=\infty$，$\lim[f(x)]^{g(x)}$ 为 1^∞

转化：$A=e^{\ln A}$，$y=[f(x)]^{g(x)}$

$\ln y=g(x)\cdot\ln f(x)$ $\Rightarrow\infty\cdot 0$ 型

取倒数化为基础型

$y=e^{\ln y}$

⑤、0^0 型

$\lim f(x)=0^+$，$\lim g(x)=0$，$\lim[f(x)]^{g(x)}$ 为 0^0

转化：$A=e^{\ln A}$，$y=[f(x)]^{g(x)}$

$\ln y=g(x)\cdot\ln f(x)$ $\Rightarrow 0\cdot(-\infty)$

化为基础型

$y=e^{\ln y}$

⑥ ∞^0 型

$\lim f(x)=\infty$，$\lim g(x)=0$，$\lim[f(x)]^{g(x)}$ 为 ∞^0

基本一样

summary：$\infty\cdot 0$ 取倒数 → $\frac{0}{0}$ or $\frac{\infty}{\infty}$

$\infty-\infty$ 通分/有理化 → $\frac{0}{0}$ or $\frac{\infty}{\infty}$

1^∞，0^0，∞^0 取对数 → $\infty\cdot 0$ 取倒数 → $\frac{0}{0}$ 或 $\frac{\infty}{\infty}$

其他 Theorems：

1. 拉格朗日中值定理（Lagrange Mean Value Theorem）

- **核心条件**：函数$f(x)$在闭区间$[a,b]$上连续，在开区间(a,b)内可导。
- **核心结论**：存在$c \in (a,b)$，使得$f'(c) = \frac{f(b)-f(a)}{b-a}$（函数在区间内某点的瞬时变化率等于区间平均变化率）。

2. 罗尔定理（Rolle's Theorem）

- **核心条件**：函数$f(x)$在闭区间$[a,b]$上连续，在开区间(a,b)内可导，且$f(a)=f(b)$。
- **核心结论**：存在$c \in (a,b)$，使得$f'(c)=0$（区间内存在导数为0的点）。

3. 柯西中值定理（Cauchy's Mean Value Theorem）

- **核心条件**：函数$f(x)$、$g(x)$在闭区间$[a,b]$上连续，在开区间(a,b)内可导，且$g'(x) \neq 0$（$\forall x \in (a,b)$）。
- **核心结论**：存在$c \in (a,b)$，使得$\frac{f(b)-f(a)}{g(b)-g(a)} = \frac{f'(c)}{g'(c)}$（两函数的平均变化率之比等于某点导数之比）。

4. 中间值定理（Intermediate Value Theorem）IVT

- **核心条件**：函数$f(x)$在闭区间$[a,b]$上连续，N is between f(a) and f(b)
- **核心结论**：若$f(a)$与$f(b)$符号相反（即$f(a)\cdot f(b)<0$），则存在$c \in (a,b)$使$f(c)=0$（连续函数在区间内取遍两端点间的所有值）。

5. 极值定理（Extreme Value Theorem）EVT

- 条件：f(x) continuous on 闭区间 [a,b]
- 结论：f(x)在[a,b]上必有绝对最大+小值

一阶导数测试：

条件：

1. f(x)在[a,b]连续，在(a,b)上(除可能在c处外)可导
2. f'(c)=0 或 f'(c)不存在

结论：

1. f'(x)在c处负变正：local minimum
2. 正→负：local maximum
3. 不变号：非极值(可能是拐点)

二阶导数测试

条件：

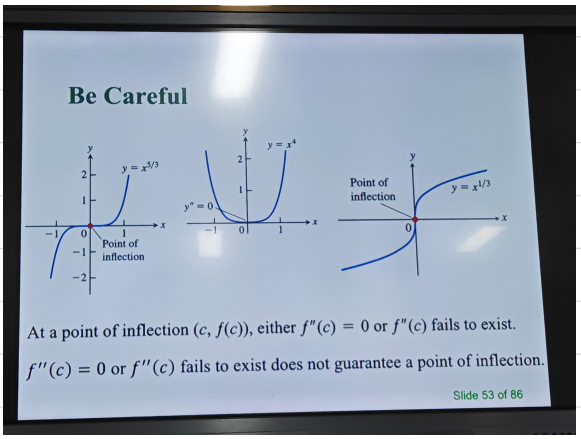
1. f(x)在包含c的区间内二阶可导，且f'(c)=0
2. f''(x)在c附近连续

结论：

1. f''(c)>0, local minimum, concave up
2. f''(c)<0, local maximum, concave down
3. f''(c)=0, use first-derivative test

拐点：

条件：c处连续，$f''(x)$ 在c处变号



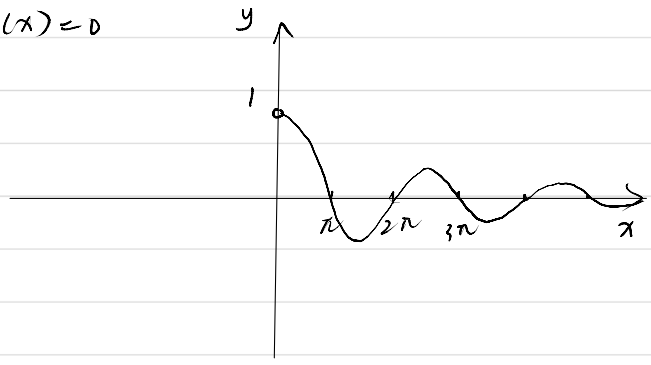
Graph sketching:

$f(x) = \frac{\sin x}{x}$

4. symmetry: even $f(x) = f(-x)$

$\lim_{x\to\infty} \frac{\sin x}{x} = 0$ $\lim_{x\to-\infty} f(x) = 0$

$\lim_{x\to 0} \frac{\sin x}{x} = 1$



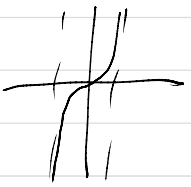
1. Domain
2. Range
3. intercepts < x / y

(increasing / decreasing + local extrema)

5. 走势 → 一阶导 6. concavity → 二阶导

$f'(x) = \frac{x\cos x - \sin x}{x^2}$ ← 恒正 → 等于 0, $x\cos x = \sin x$, $x = \tan x$

↓



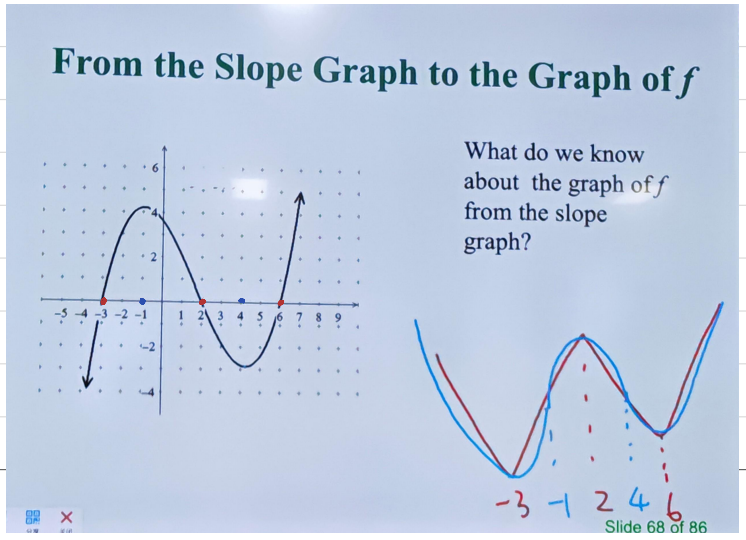
变负 ↑ 变正 ↓

$g(x) = \tan x - x$

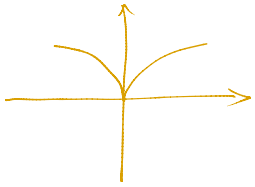
$g'(x) = \sec^2 x - 1 \geq 0$

$g(0) = 0$

$\therefore g(x) \geq 0$



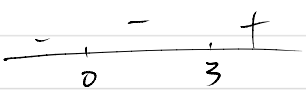
$x^{\frac{2}{3}}$ 的 graph:



$f'(x) = 4x^3 - 12x^2$, $f'(x) = 0$

$4x^3 = 12x^2$

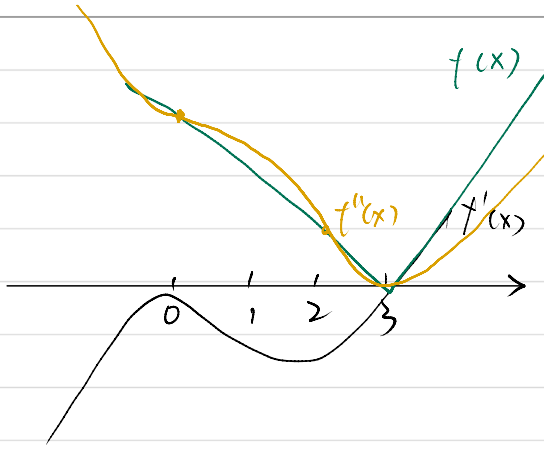
$x_1 = 0,\ x_2 = 3$



$x = 3$, local minima

$(-\infty, 3)$ decreasing

$(3, \infty)$ increasing



Optimization

结果看情况取小数/整数

Rules for Definite Integrals

1. *Zero:* $\int_a^a f(x)\,dx = 0$
2. *Additivity:* $\int_a^b f(x)\,dx + \int_b^c f(x)\,dx = \int_a^c f(x)\,dx$
3. *Order of Integration:* $\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx$
4. *Constant Multiple:* $\int_a^b kf(x)\,dx = k\int_a^b f(x)\,dx$ Any number k

   $\int_a^b -f(x)\,dx = -\int_a^b f(x)\,dx$ $k = -1$
5. *Sum and Difference:* $\int_a^b (f(x) \pm g(x))\,dx = \int_a^b f(x)\,dx \pm \int_a^b g(x)\,dx$
6. *Max-Min Inequality:* If max f and min f are the maximum and minimum values of f on $[a, b]$, then

   $$\min f \cdot (b - a) \le \int_a^b f(x)\,dx \le \max f \cdot (b - a).$$
7. *Domination:* $f(x) \ge g(x)$ on $[a, b] \Rightarrow \int_a^b f(x)\,dx \ge \int_a^b g(x)\,dx$

   $f(x) \ge 0$ on $[a, b] \Rightarrow \int_a^b f(x)\,dx \ge 0$ $g = 0$

Definition of Derivative: $\lim_{x \to h} \frac{F(x+h) - F(x)}{h}$

The Foundamental Theorem of Calculus, Antiderivative Part

If f is continuous on $[a, b]$, then the function

$$F(x) = \int_a^x f(t)\,dt$$

has a derivative at very point x in $[a, b]$, and

$$\frac{dF}{dx} = \frac{d}{dx}\int_a^x f(t)\,dt = f(x)$$

Prove:

$$\lim_{h\to 0}\frac{\int_a^{x+h} f(t)\,dt - \int_a^{x} f(t)\,dt}{h} = \lim_{h\to 0}\frac{\int_x^{x+h} f(t)\,dt}{h}$$

$f_{min}\cdot(b-a) \le \int_a^b f(x)\,dx \le f_{max}\cdot(b-a)$ [extreme value theorem]

$$f(w)\cdot h \le \int_x^{x+h} f(t)\,dt \le f(v)\cdot h$$

$$\frac{\int_x^{x+h} f(t)\,dt}{h}$$

$$\lim_{h\to 0} f(w) \le \lim_{h\to 0}\frac{\int_x^{x+h} f(t)\,dt}{h} \le \lim_{h\to 0} f(v)$$

[squeeze theorem] $\Downarrow$

$f(x)$

Example:

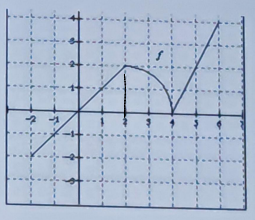
$f(x) = \int_1^x (-t^2 - t + 6)\,dt$, intervals of increasing f

$f'(x) = -x^2 - x + 6$

$f'(x) > 0$

$-3 < x < 2$

Let $F(x) = \int_2^x f(t)\,dt$. The graph of f on the interval $[-2,6]$ consists of two line segments and a quarter of a circle, as shown at right.



(a) Find $F(0)$ and $F(4)$.

$-2 \qquad \pi$

(b) Determine the interval where $F(x)$ is increasing. Justify your answer.

$x \in [0,4) \cup (4,6]$

(c) Find the critical numbers of $F(x)$ and determine if each corresponds to a relative minimum value, a relative maximum value, or neither. Justify your answers.

$x = 0$ minimum

(d) Find the absolute extreme values of $F(x)$ and the x-values at which they occur. Justify your answers. $-2 \quad \pi \quad 0 \quad \pi+4$

$F(0), F(4), F(-2), F(6)$

(e) Find the x-coordinates of the inflection points of $F(x)$. Justify your answer.

$f'(x) < 0$ undefined $\rightarrow x = 2, x = 4$

(f) Determine the intervals where the graph of $F(x)$ is concave down. Justify your answer.

The Derivative of the Accumulation Function with the Chain Rule

If $F(x)=\int_a^x f(t)\,dt$, where a is constant and f is a continuous function, then:

$$F'(x)=f(x).$$

If $F(x)=\int_a^{g(x)} f(t)\,dt$, where a is constant, f is a continuous function, and g is a differentiable function, then:

$$F'(x)=f\big(g(x)\big)\cdot g'(x).$$

$y=\int_x^5 3t\sin t\,dt$

$\Rightarrow y=-\int_5^x 3t\sin t\,dt$

$y'=-3x\sin x$

$y=\int_{2x}^{x^2}\frac{1}{2+e^t}$

$\Rightarrow y=\int_{2x}^{0}\frac{1}{2+e^t}dt+\int_0^{x^2}\frac{1}{2+e^t}dt$

$=-\int_0^{2x}\frac{1}{2+e^t}dt+\int_0^{x^2}\frac{1}{2+e^t}dt$

$y'=-\frac{1}{2+e^{2x}}\cdot 2+\frac{1}{2+e^{x^2}}\cdot 2x$

The Fundamental Theorem of Calculus, Evaluation Part.

$F'(x)=f(x)$

$\int_a^b f(x)\,dx=F(b)-F(a)$

Example:

$f(x)=\frac{|x|(x^2-1)}{x}$, $\int_{-1}^{2} f(x)\,dx=?$

$\int_{-1}^{0} f(x)\,dx+\int_0^2 f(x)\,dx$

$=\int_{-1}^{0}(1-x^2)\,dx+\int_0^2(x^2-1)\,dx$

$=0-(\frac{1}{3}-1)+\frac{8}{3}-2$

$=\frac{4}{3}$

The integral of a Rate is the Net Change

displacement = $\int_0^8 (49 - 9.8t)\,dt$

total distance: $\int_0^8 |49 - 9.8t|\,dt$

Indefinit Integral

$\int f(x)\,dx = F(x) + C$

Formulas

Properties of Indefinite Integrals

$$\int k\,f(x)\,dx = k\int f(x)\,dx \quad \text{for any constant } k$$

$$\int (f(x) \pm g(x))\,dx = \int f(x)\,dx \pm \int g(x)\,dx$$

Power Formulas

$$\int u^n\,du = \frac{u^{n+1}}{n+1} + C \text{ when } n \neq -1$$

$$\int u^{-1}\,du = \int \frac{1}{u}\,du = \ln|u| + C$$

Trigonometric Formulas

$$\int \cos u\,du = \sin u + C$$

$$\int \sin u\,du = -\cos u + C$$

$$\int \sec^2 u\,du = \tan u + C$$

$$\int \csc^2 u\,du = -\cot u + C$$

$$\int \sec u \tan u\,du = \sec u + C$$

$$\int \csc u \cot u\,du = -\csc u + C$$

Exponential and Logarithmic Formulas

$$\int e^u\,du = e^u + C$$

$$\int a^u\,du = \frac{a^u}{\ln a} + C$$

Integrating using substitution

Example:

a. $\int 8x\sqrt{4x^2+1}\,dx$

Let $u = 4x^2+1$

$du = 8x\,dx$

$I = \int \sqrt{u}\,du$

$= \frac{u^{\frac{3}{2}}}{\frac{1}{2}+1} + C$

$= \frac{2}{3}(4x^2+1)^{\frac{3}{2}} + C$

b. $\int x(x^2+2)^3\,dx$

Let $u = x^2+2$

$du = 2x\,dx$

$x\,dx = \frac{1}{2}\,du$

$I = \int u^3 \cdot \frac{1}{2}\,du$

$= \frac{1}{2}\int u^3\,du$

$= \frac{1}{2}\cdot\frac{u^4}{4} + C$

$= \frac{1}{8}(x^2+2)^4 + C$

c. $\int \frac{\ln x}{x}\,dx$

Let $u = \ln x$

$du = \frac{1}{x}\,dx$

$I = \int u\,du = \frac{u^2}{2} + C = \frac{(\ln x)^2}{2} + C$

d. $\int (3x^2+1)(e^{x^3}\cdot e^x)\,dx$

$= \int (3x^2+1)(e^{x^3+x})\,dx$

Let $u = x^3+x$

$du = 3x^2+1\,dx$

$I = \int e^u\,du$

$= e^{x^3+x} + C$

e. $\int \frac{e^{3t}}{3+e^{3t}}\,dt$

Let $u = 3+e^{3t}$

$du = 3e^{3t}\,dt$

$I = \int \frac{1}{3u}\,du$

$= \frac{1}{3}\cdot\ln(3+e^{3t}) + C$

f. $\int \cos^4 x \sin x\,dx$

Let $u = \cos x$

$du = -\sin x\,dx$

$I = -\int u^4\,du$

g. $\int (x\cos x^2)(\sin x^2+5)\,dx$

Let $u = \sin x^2+5$

$du = 2x\cos x^2\,dx$

$x\cos x^2\,dx = \frac{1}{2}\,du$

$I = \frac{1}{2}\int u\,du$

h. $\int \sin x\, e^{\cos x}\,dx$

Let $u = \cos x$

$du = -\sin x\,dx$

$I = \int -e^u\,du$

$= -e^{\cos x} + C$

i. $\int \cot 7x\,dx$

Let $u = \sin 7x$

$du = 7\cos 7x\,dx$

$I = \frac{1}{7}\int \frac{1}{u}\,dx$

$= \frac{1}{7}\ln|u| + c$

$= \frac{1}{7}\ln|\sin 7x| + c$

j. $\int \cot^2 3x\,dx$

$= \int (\csc^2 3x - 1)\,dx$

$= -\frac{1}{3}\cot 3x - \frac{x^2}{2}$

Substitution in Definite Integrals.

$$\int_a^b f(g(x))\,g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du$$

numerator's degree is less than the denominator

a. $\int \frac{dx}{5+4x^2} \Rightarrow \int \frac{1}{a^2+x^2}\,dx$

b. $\int \frac{1}{\sqrt{4-x^2}}\,dx \Rightarrow \int \frac{1}{\sqrt{a^2-x^2}}\,dx$

$$\int \frac{dx}{1+x^2} = \tan^{-1}x + c$$

$$\int \frac{1}{\sqrt{1-x^2}}\,dx = \sin^{-1}x + c$$

$I = \int \frac{1}{5+(2x)^2}\,dx$

$= \int \frac{1}{5\left(1+\frac{(2x)^2}{5}\right)}\,dx$

$= \frac{1}{5}\int \frac{1}{1+\left(\frac{2x}{\sqrt{5}}\right)^2}\,dx$

let $u = \frac{2}{\sqrt{5}}x$

$du = \frac{2}{\sqrt{5}}\,dx$

$I = \frac{1}{5}\cdot\frac{\sqrt{5}}{2}\int \frac{1}{1+u^2}\,du$

$= \frac{\sqrt{5}}{10}\tan^{-1}u + c$

$= \frac{\sqrt{5}}{10}\tan^{-1}\left(\frac{2}{\sqrt{5}}x\right) + c$

$I = \int \frac{1}{\sqrt{4\left(1-\left(\frac{x}{2}\right)^2\right)}}\,dx$

$= \frac{1}{2}\int \frac{1}{\sqrt{1-\left(\frac{x}{2}\right)^2}}\,dx$

let $u = \frac{x}{2}$

$du = \frac{1}{2}\,dx$

$I = \int \frac{1}{\sqrt{1-u^2}}\,du$

$= \sin^{-1}u + c$

$= \sin^{-1}\frac{x}{2} + c$

(practice)

c. $\int \frac{1}{x^2+12x+37}\,dx$ ⇒ completing the square 配方

$I = \int \frac{1}{(x+6)^2+1}\,dx$

let $u = x+6$

$du = du$

$I = \int \frac{1}{u^2+1}\,du$

$= \tan^{-1}u + C$

$= \tan^{-1}(x+6) + C$

(practice)

d. $\int \frac{1}{\sqrt{4-6x-9x^2}}\,dx$

$I = \int \frac{1}{\sqrt{4-[(3x+1)^2-1]}}\,dx$

$= \int \frac{1}{\sqrt{5-(3x+1)^2}}\,dx$

$= \frac{1}{\sqrt{5}}\int \frac{1}{\sqrt{1-\left(\frac{3x+1}{\sqrt{5}}\right)^2}}\,dx$

let $u = \frac{3x+1}{\sqrt{5}}$

$du = \frac{3}{\sqrt{5}}\,dx$

$= \frac{1}{\sqrt{5}} \cdot \frac{\sqrt{5}}{3}\int \frac{1}{\sqrt{1-u^2}}\,du$

$= \frac{1}{3}\int \frac{1}{\sqrt{1-u^2}}\,du$

$= \frac{1}{3}\sin^{-1}u + C$

$= \frac{1}{3}\sin^{-1}\left(\frac{3x+1}{\sqrt{5}}\right) + C$

e. $\int \frac{8}{\sqrt{12-x^2-4x}}\,dx$

$I = \int \frac{8}{\sqrt{16-(x+2)^2}}\,dx$

$= \frac{1}{4}\int \frac{8}{\sqrt{1-\left(\frac{x+2}{4}\right)^2}}\,dx$

$= 2\int \frac{1}{\sqrt{1-\left(\frac{x+2}{4}\right)^2}}\,dx$

let $u = \frac{x+2}{4}$

$du = \frac{1}{4}\,dx$

$I = 8\int \frac{1}{\sqrt{1-u^2}}\,du$

$= 8\sin^{-1}u + C$

$= 8\sin^{-1}\left(\frac{x+2}{4}\right) + C$

f. $\int \frac{1}{x^2-1}\,dx$ => factorization => partial fraction

$I = \int \frac{1}{(x+1)(x-1)}\,dx$

$= \int \frac{A(x-1)+B(x+1)}{(x+1)(x-1)}\,dx$

$A(x-1)+B(x+1)=1$

let $A(x-1)=0$

$x=1$

$2B=1$

$B=\frac{1}{2}$

let $B(x+1)=0$

$x=-1$

$-2A=1$

$A=-\frac{1}{2}$

$I = \int \left(-\frac{1}{2}\cdot\frac{1}{x+1}+\frac{1}{2}\cdot\frac{1}{x-1}\right)dx$

$= -\frac{1}{2}\ln|x+1| + \frac{1}{2}\ln|x-1| + C$

g. $\int \frac{x^2+2x-1}{2x^3+3x^2-2x}\,dx$

$I = \int \frac{x^2+2x-1}{x(2x-1)(x+2)}\,dx$

$= \int \left(\frac{A}{x}+\frac{B}{2x-1}+\frac{C}{x+2}\right)dx$

$= \int \frac{A(2x-1)(x+2)+Bx(x+2)+Cx(2x-1)}{x(2x-1)(x+2)}\,dx$

let $x=-2$, $4-4-1=C(-2)(-5)$

$C=-\frac{1}{10}$

let $x=\frac{1}{2}$, $\frac{1}{4}+1-1=\frac{1}{2}\cdot\frac{5}{2}B$

$B=\frac{1}{5}$

let $x=0$, $A=\frac{1}{2}$

$I = \int \left(\frac{\frac{1}{2}}{x}+\frac{\frac{1}{5}}{2x-1}-\frac{\frac{1}{10}}{x+2}\right)dx$

$= \frac{1}{2}\int\frac{1}{x}\,dx + \frac{1}{5}\int\frac{1}{2x-1}\,dx - \frac{1}{10}\int\frac{1}{x+2}\,dx$

$= \frac{1}{2}\ln|x| + \frac{1}{5}\cdot\frac{1}{2}\ln|2x-1| - \frac{1}{10}\ln|x+2| + C$

Improper Rational Functions - Long division

a. $\int \frac{3x^3+8x^2-10x+5}{x+4}\,dx$

$= \int \left(3x^2 - 4x + 6 - \frac{9}{x+4}\right)dx$ $\Rightarrow$ Long Division

b. $\int \frac{x^3+x}{x-1}\,dx$

$= \int \left(x^2+x+2+\frac{2}{x-1}\right)dx$

c. $\int \frac{12x^2}{2x+1}\,dx$

$= \int \left(6x - \frac{6x}{2x+1}\right)dx$

d. $\int \frac{4x^3}{2x+3}\,dx$

$= \int \left(2x^2 - 3x + \frac{9}{2} - \frac{27}{4x+6}\right)dx$

e. $\int_3^5 \frac{2x^2+x+4}{x-1}\,dx$

$= \int_3^5 \left(2x+3+\frac{7}{x-1}\right)dx$

$= \left[x^2+3x+7\ln|x-1|\right]_3^5$

Integration by parts

a. $\int xe^x\,dx$

$(uv)' = u'v + uv'$

$\int (uv)'\,dx = \int (u'v + uv')\,dx$

$uv = \int u'v\,dx + \int uv'\,dx$

$u = x$, $v = e^x$

$= \int 1 \cdot e^x\,dx + \int xe^x\,dx$

$= e^x + \int xe^x\,dx$

$\int xe^x\,dx = uv - e^x + c$

$= xe^x - e^x + c$

b. $\int x\cos x\,dx$

① $uv = \int u'v\,dx + \int uv'\,dx$

② $\int uv'\,dx = uv - \int u'v\,dx$

$\int u\,dv = uv - \int v\,du$

③ $\int x\cos x\,dx$

Let $u = x$, $v' = \cos x$

$u' = 1$, $v = \sin x$

$= x\sin x - \int \sin x\,dx$

$= x\sin x + \cos x + c$

$\int \tan^{-1} y \, dy$

c. $\int x^2 e^x dx$

$\int u dv = uv - \int v du$

let $u = x^2$ $v' = e^x$

$u' = 2x$ $v = e^x$

$I = x^2 e^x - \int 2x e^x dx$

$= x^2 e^x - 2\int x e^x dx$

$= x^2 e^x - 2x e^x + 2e^x + C$

d. $\int x \sin \frac{x}{2} dx$

$\int u dv = uv - \int v du$

let $u = x$ $v' = \sin \frac{x}{2}$

$u' = 1$ $v = -2\cos \frac{x}{2}$

$I = -2x\cos \frac{x}{2} - \int (-2\cos \frac{x}{2}) dx$

$= -2x\cos \frac{x}{2} + 2\int \cos \frac{x}{2} dx$

$= -2x\cos \frac{x}{2} + 4\sin \frac{x}{2} + C$

e. $\int t^2 \cos t \, dt$

$\int u dv = uv - \int v du$

let $u = t^2$, $v' = \cos t$

$u' = 2t$, $v = \sin t$

$I = t^2 \sin t - \int 2t \sin t \, dt$

$= t^2 \sin t - 2\int t \sin t \, dt$

$\int t \sin t \, dt$

let $u = t$, $v' = \sin t$

$u' = 1$, $v = -\cos t$

$= -t\cos t - \int -\cos t \, dt$

$= -t\cos t + \sin t + C$

$= t^2 \sin t + 2t\cos t - 2\sin t + C$

f. $\int \ln x \, dx$

$= \int 1 \cdot \ln x \, dx$

let $u = \ln x$, $v' = 1$

$u' = \frac{1}{x}$, $v = x$

$= x\ln x - \int \frac{1}{x} x \, dx$

$= x\ln x - x + C$

g. $\int \sin^{-1} x \, dx$

$= \int 1 \cdot \sin^{-1} x \, dx$

let $u = \sin^{-1} x$ $v' = 1$

$u' = \frac{1}{\sqrt{1-x^2}}$ $v = x$

$= x \cdot \sin^{-1} x - \int \frac{x}{\sqrt{1-x^2}} dx$

$\int \frac{x}{\sqrt{1-x^2}} dx$

let $u = 1 - x^2$ $du = -2x \, dx$

$\int \frac{x}{\sqrt{1-x^2}} dx = -\frac{1}{2}\int \frac{1}{\sqrt{u}} du$

$= -\frac{1}{2} \cdot 2u^{\frac{1}{2}} + C$

$= -u^{\frac{1}{2}} + C$

$= -(1-x^2)^{\frac{1}{2}} + C$

$= x \cdot \sin^{-1} x + (1-x^2)^{\frac{1}{2}} + C$

h. $\int \tan^{-1} x \, dx$

$= \int 1 \cdot \tan^{-1} x \, dx$

let $u = \tan^{-1} x$ $v' = 1$

$u' = \frac{1}{1+x^2}$ $v = x$

$I = x\tan^{-1} x - \int \frac{x}{1+x^2} dx$

$\int \frac{x}{1+x^2} dx$

let $u = 1 + x^2$ $du = 2x \, dx$

$\frac{1}{2}\int \frac{1}{u} du$

$= \frac{1}{2}\ln|u| + C$

$= \frac{1}{2}\ln|1+x^2| + C$

$= x\tan^{-1} x - \frac{1}{2}\ln|1+x^2| + C$

Repeated Integration by parts

a. if $\int_2^5 f(x)\,dx = 14$, $\int_2^5 x \cdot f'(x)\,dx = ?$

x	2	5
$f(x)$	4	7
$f'(x)$	2	3

let $u = x$ $v' = f'(x)$

$u' = 1$ $v = f(x)$

$$\int_2^5 x \cdot f'(x)\,dx = [x f(x)]_2^5 - \int_2^5 f(x)\,dx$$

$$= 35 - 8 - 14$$

$$= 13$$

b. $\int e^x \cos x\,dx$

let $u = e^x$ $v' = \cos x$

$u' = e^x$ $v = \sin x$

$I = e^x \sin x - \int e^x \sin x\,dx$

let $u = \cos x$ $v' = e^x$

$u' = -\sin x$ $v = e^x$

$I = e^x \cos x + \int e^x \sin x\,dx$

$I = (e^x \sin x - \int e^x \sin x\,dx + e^x \cos x + \int e^x \sin x\,dx)\frac{1}{2}$

$= \frac{1}{2} e^x (\sin x + \cos x)$

$\int e^x \sin x\,dx$

let $u = e^x$ $v' = \sin x$

$u' = e^x$ $v = -\cos x$

$\int e^x \sin x\,dx = -e^x \cos x + \int e^x \cos x\,dx$

$I = e^x \sin x + e^x \cos x - \int e^x \cos x\,dx$

$2I = e^x \sin x + e^x \cos x$

$I = \frac{1}{2} e^x (\sin x + \cos x)$

Improper integrals

1. $\int_a^{\infty} f(x)\,dx = \lim_{b\to\infty} \int_a^b f(x)\,dx$

2. $\int_{-\infty}^{b} f(x)\,dx = \lim_{a\to-\infty} \int_a^b f(x)\,dx$

3. $\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{c} f(x)\,dx + \int_c^{\infty} f(x)\,dx$

limit exist → integral converges

a. $\int_1^{\infty} \frac{\ln x}{x^2}\,dx$

$= \lim_{b\to\infty} \int_1^b \frac{\ln x}{x^2}\,dx$

let $u = \ln x \quad v' = \frac{1}{x^2}$

$u' = \frac{1}{x} \quad v = -\frac{1}{x}$

$I = \lim_{b\to\infty} \left[-\frac{1}{x}\ln x\right]_1^b - \int_1^b \left(-\frac{1}{x^2}\right) dx$

$= \lim_{b\to\infty} -\frac{\ln b}{b} - \left[\frac{1}{x}\right]_1^b$

$= \lim_{b\to\infty} -\frac{\ln b}{b} - \left(\frac{1}{b} - 1\right)$

$= \lim_{b\to\infty} \left(-\frac{\ln b}{b} - \frac{1}{b}\right) + 1$

$= \lim_{b\to\infty} \left(\frac{-\frac{1}{b}}{1} - 0\right) + 1$ ← L'Hospital Rule

$= 1$

b. $\int_1^{\infty} \frac{1}{x}\,dx$

$= \lim_{b\to\infty} \int_1^b \frac{1}{x}\,dx$

$= \lim_{b\to\infty} \ln|b| - \ln 1$

$= \lim_{b\to\infty} \ln|b|$

$= \infty$

$\therefore \infty$ is diverge.

$\int_1^{\infty} \frac{1}{x^p}\,dx = \int_1^{\infty} x^{-p}\,dx$

$= \lim_{b\to\infty} \int_1^b x^{-p}\,dx$

$= \lim_{b\to\infty} \frac{x^{1-p}}{1-p}\Big]_1^b$

$= \lim_{b\to\infty} \left(\frac{b^{1-p}}{1-p} - \frac{1}{1-p}\right)$

if $p<1$, $I=\infty$, diverge

if $p>1$, $I=\frac{1}{p-1}$, converge

两个improper相加，
一个趋向无穷
就是diverge，
$\infty - \infty$ 也是diverge

Integrals of functions that become infinite at a point

1. continuous on $[a, b]$, discontinuous at a

$$\int_a^b f(x)\,dx = \lim_{c\to a^+} \int_c^b f(x)\,dx$$

2. continuous on $[a, b)$ and discontinuous at b

$$\int_a^b f(x)\,dx = \lim_{c\to b^-} \int_a^c f(x)\,dx$$

3. discontinuous at c, continuous at $[a,c)\cup(c,b]$

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$

a. $\int_1^2 \frac{1}{(x-2)}\,dx$

$= \lim_{b\to 2^-} \int_1^b \frac{dx}{x-2}$

$= \lim_{b\to 2^-} \ln|x-2| \Big]_1^b$

$= \lim_{b\to 2^-} (\ln|b-2| - \ln 1)$

$= \lim_{b\to 2^-} \ln|b-2|$

$= -\infty$

b. $\int_0^3 \frac{dx}{(x-1)^{\frac{2}{3}}}$

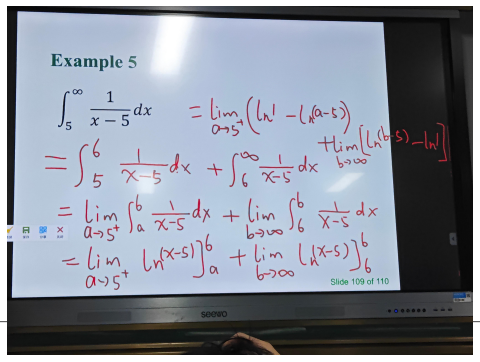
$= \int_0^1 \frac{dx}{(x-1)^{\frac{2}{3}}}\,dx + \int_1^3 \frac{dx}{(x-1)^{\frac{2}{3}}}\,dx$

$= \lim_{b\to 1^-} \int_0^b \frac{dx}{(x-1)^{\frac{2}{3}}}\,dx + \lim_{a\to 1^+} \int_a^3 \frac{dx}{(x-1)^{\frac{2}{3}}}\,dx$

$= \lim_{b\to 1^-} 3(x-1)^{\frac{1}{3}} \Big]_0^b + \lim_{a\to 1^+} 3(x-1)^{\frac{1}{3}} \Big]_a^3$

$= \lim_{b\to 1^-} \left[3(b-1)^{\frac{1}{3}} - 3(0-1)^{\frac{1}{3}}\right] + \lim_{a\to 1^+} \left[3(2)^{\frac{1}{3}} - 3(a-1)^{\frac{1}{3}}\right]$

$= 3\sqrt[3]{2} + 3$



d. $\int_1^\infty xe^{-x}\,dx$

$= \lim_{b\to\infty} \int_1^b xe^{-x}\,dx$ Let $u = x$ $v' = e^{-x}$
$u' = 1$ $v = -e^{-x}$

$= \lim_{b\to\infty} \left[-xe^{-x}\right]_1^b - \int_1^b -e^{-x}\,dx$

$= \lim_{b\to\infty} \left[-xe^{-x} - e^{-x}\right]_1^b$

$= \lim_{b\to\infty} (-be^{-b} - e^{-b} + e^{-1} + e^{-1})$

$= 2e^{-1} = \frac{2}{e}$

e. $\int_0^\infty \frac{2}{x^2+4x+3}\,dx$

$= 2\int_0^\infty \frac{1}{(x+1)(x+3)}\,dx$

$= 2\lim_{b\to\infty}\int_0^b \frac{1}{(x+1)(x+3)}\,dx$

$A(x+3)+B(x+1)=1$

let $x=-1$

$A=\frac{1}{2}$

let $x=-3$

$B=-\frac{1}{2}$

$I=\lim_{b\to\infty}\left[\left(\frac{1}{2}\int\frac{1}{x+1}\,dx-\frac{1}{2}\int\frac{1}{x+3}\,dx\right)\right]_0^b$

$=\lim_{b\to\infty}\left[\ln|x+1|-\ln|x+3|\ \right]_0^b$

$=\lim_{b\to\infty}\left(\ln|b+1|-\ln|b+3|-\ln 1+\ln 3\right)$

$=\lim_{b\to\infty}\left(\ln\left|\frac{b+1}{b+3}\right|+\ln 3\right)$

$=\lim_{b\to\infty}\left(\ln 1+\ln 3\right)$

$=\ln 3$

f. $\int_0^1 \frac{1}{1-x}\,dx$

$=\lim_{b\to 1^-}\int_0^b \frac{1}{1-x}\,dx$

$=\lim_{b\to 1^-}\left[\ln|1-x|\right]_0^b$

$=\lim_{b\to 1^-}\left(\ln|1-b|-\ln 1\right)$

$=\lim_{b\to 1^-}\ln(1-b)$

$=-\infty$

$\because -\infty$ is diverge

Unit 7

Differential equations

a. $\frac{dy}{dx} = \sec^2 x + 2x + 5$

general solution: a family of solutions

b. $\frac{dy}{dx} = e^x - 6x^2$, through point $(1, 0)$

$y = e^x - 2x^3 + C$

$0 = e - 2 + C$

$C = 2 - e$

$y = e^x - 2x^3 + 2 - e$ particular solution: only 1 solution

c. $k = ?$ so $y = k\sin(5x) + 2\cos(4x)$ is a solution of

$y'' + 16y = -27\sin(5x)$

$y' = 5k\cos(5x) - 8\sin(4x)$ => first derivative

$y'' = -25k\sin(5x) - 32\cos(4x)$ => second derivative

$-25k\sin(5x) - 32\cos(4x) + 16k\sin(5x) + 32\cos(4x)$

$= -9k\sin(5x)$

$k = 3$

d. which are the solutions to $y'' - 10y' + 9y = 0$

✗ I. $y = 2\sin(3x)$

✓ II. $y = 5e^x$

✓ III. $y = Ce^{9x}$, c = constant

Practice 1

1. Find the general solution to the differential equation

$\frac{dy}{dx} = 5x^4 - \sec^2 x$

$\frac{dy}{dx} = \sin x - e^{-x} + 8x^3$

$\frac{dy}{dx} = \frac{1}{\sqrt{1 - x^2}} - \frac{1}{\sqrt{x}}$

2. Solve the initial value problem.

$\frac{dy}{dt} = \frac{1}{1 + t^2} + 2^t \ln 2$ and $y = 3$ when $t = 0$

Approximating Solutions using Euler's Method

Euler's Method for Graphing a Solution to an Initial Value Problem

1. Begin at the point (x, y) specified by the initial condition. This point will be on the graph, as required.
2. Use the differential equation to find the slope dy/dx at the point.
3. Increase x by a small amount, Δx. Increase y by a small amount, Δy, where $\Delta y = (dy/dx)\Delta x$. This defines a new point $(x + \Delta x, y + \Delta y)$ that lies along the linearization (Figure 7.7).
4. Using this new point, return to step 2. Repeating the process constructs the graph to the right of the initial point.
5. To construct the graph moving to the left from the initial point, repeat the process using negative values for Δx.

a. $\frac{dy}{dx} = 2x - y$, if $y=3$, $x=2$, five equal steps, y? when $x=1.5$

(x_0, y_0)	Δx	$\frac{dy}{dx}$	Δy	(x_i, y_i)
$(2, 3)$	-0.1	1	-0.1	$(1.9, 2.9)$
$(1.9, 2.9)$	-0.1	0.9	-0.09	$(1.8, 2.81)$
$(1.8, 2.81)$	-0.1	0.79	-0.079	$(1.7, 2.731)$
$(1.7, 2.731)$	-0.1	0.669	-0.0669	$(1.6, 2.6641)$
$(1.6, 2.6641)$	-0.1	0.5359	-0.05359	$(1.5, 2.61051)$

Seperable Differential Equation

$\frac{1}{f(x)}dy = g(x)dx \Rightarrow$ seperate variables

a. $\frac{dy}{dx} = 2xy$

$\frac{dy}{y} = 2x\,dx$

$\int \frac{dy}{y} = \int 2x\,dx$

$\ln|y| = x^2 + C$

$|y| = e^{x^2} + C$

$y = \pm e^{x^2} + C$

b. $\frac{dy}{dx} = (\cos x)\, e^{y+\sin x}$

$\frac{dy}{e^y} = (\cos x)\, e^{\sin x}\,dx$

$\int \frac{dy}{e^y} = \int \cos x \cdot e^{\sin x}\,dx$ → add integral

$u = \sin x \quad du = \cos x$

$\int \cos x \cdot e^{\sin x}\,dx = \int e^u\,du = e^u + C = e^{\sin x} + C$

$-e^{-y} = e^{\sin x} + C$

$-e^{-y} = -e^{\sin x} + C$

$-y = \ln|-e^{\sin x} + C|$

$y = -\ln|-e^{\sin x} + C|$

c. $\frac{dy}{dx} = (xy)^2$, $y=1$, $x=1$

$\frac{1}{y^2}dy = x^2dx$

$\int \frac{1}{y^2}dy = \int x^2dx$

$-\frac{1}{y} = \frac{x^3}{3} + c$

when $y=1$, $x=1$ $\Rightarrow$ initial condition

$-1 = \frac{1}{3} + c$

$c = -\frac{4}{3}$ $\Rightarrow$ find value of c

$-\frac{1}{y} = \frac{x^3}{3} - \frac{4}{3}$

$y = \frac{3}{4-x^3}$

d. $\frac{dy}{dx} = \frac{1+x}{xy}$, $y(1) = -4$

$y\,dy = \frac{1+x}{x}dx$

$\int y\,dy = \int \frac{1}{x} + 1\,dx$

$\frac{1}{2}y^2 = \ln|x| + x + c$

$\frac{1}{2}\cdot 16 = 0 + 1 + c$

$c = 7$

$\frac{1}{2}y^2 = \ln|x| + x + 7$

$y^2 = 2(\ln|x| + x + 7)$

$y = \pm\sqrt{2(\ln|x| + x + 7)}$ $\Rightarrow$ find the sign

$y = \pm\sqrt{16}$

$\because y(1) = -4$ $\Rightarrow$ initial condition

$\therefore y = -\sqrt{2(\ln|x| + x + 7)}$

e. $\frac{dy}{dx} = \frac{y-1}{x^2}$, find $y = f(x)$, $f(2) = 0$

$\frac{1}{y-1}dy = \frac{1}{x^2}dx$

$\int \frac{1}{y-1}dy = \int \frac{1}{x^2}dx$

$\ln|y-1| = -\frac{1}{x} + c$

$\because f(2) = 0$

$0 = -\frac{1}{2} + c$

$c = \frac{1}{2}$

$|y-1| = e^{-\frac{1}{x} + \frac{1}{2}}$

$y = 1 \pm e^{-\frac{1}{x} + \frac{1}{2}}$

$\because f(2) = 0$

$\therefore y = 1 - e^{-\frac{1}{x} + \frac{1}{2}}$

(practice)

d. $\frac{dy}{dx} = (y+5)(x+2)$, $f(0)=1$

$\frac{1}{y+5}dy = (x+2)\,dx$

$\int \frac{1}{y+5}dy = \int (x+2)\,dx$

$\ln|y+5| = \frac{1}{2}x^2+2x+C$

$\because f(0)=1$

$C = \ln 6$

$\ln|y+5| = \frac{1}{2}x^2+2x+\ln 6$

$\ln\left|\frac{y+5}{6}\right| = \frac{1}{2}x^2+2x$

$\left|\frac{y+5}{6}\right| = e^{\frac{1}{2}x^2+2x}$

$y = \pm 6e^{\frac{1}{2}x^2+2x} - 5$

$\because f(0)=1$

$\therefore y = 6e^{\frac{1}{2}x^2+2x} - 5$

(practice)

e. $\frac{dy}{dx} = \cos^2 y$, $f(0)=0$

$\frac{1}{\cos^2 y}dy = 1\,dx$

$\int \frac{1}{\cos^2 y}dy = \int 1\,dx$

$\int \sec^2 y\,dy = \int 1\,dx$

$\tan^{-1} y = x + C$

$\because f(0)=0$

$C = 0$

$\tan^{-1} y = x$

$y = \tan x$

(practice)

f. $\frac{dy}{dx} = -2xy^2$, $f(1)=0.25$

$\frac{1}{y^2}dy = -2x\,dx$

$\int y^{-2}dy = \int -2x\,dx$

$-y^{-1} = -x^2 + C$

$\because f(1)=0.25$

$-4 = -1 + C$

$C = -3$

$-y^{-1} = -x^2 - 3$

$y = \frac{1}{x^2+3}$

logistic regretion

increase at a rate proportional to the number present:

$\frac{dy}{dx} = ky$

$y = e^{kx+c}$

$y = c \cdot e^{kx}$

$(y_0 = c)$

logistic growth Model

$\frac{dP}{dt} = kP(M-P)$

solution: $P = \frac{M}{1+Ae^{-Mkt}}$

M = carrying capacity

k = logistic growth constant

Unit 8

Average value

$$av(f) = \frac{1}{b-a}\int_a^b f(x)\,dx$$

$$F(b)-F(a) = \int_a^b F'(x)\,dx$$
$$F(b) = F(a) + \int_a^b F'(x)\,dx$$

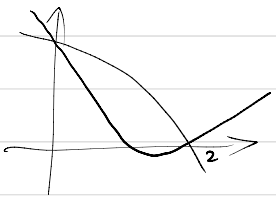
Area between curves

$$A = \int_a^b [f(x) - g(x)]\,dx$$ (top curve)

a. $y = e^x + 2$, $y = \frac{1}{1+x^2}$

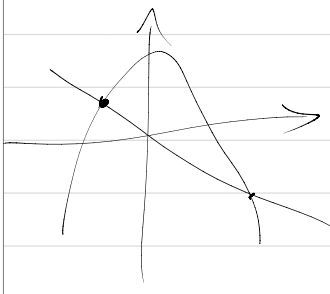
$$A = \int_0^1 \left[e^x + 2 - \frac{1}{1+x^2}\right] dx$$
$$= \left[e^x + 2x - \tan^{-1}x\right]_0^1$$
$$= \left(e + 2 - \frac{\pi}{4}\right) - 1$$
$$= e + 1 - \frac{\pi}{4}$$

b. $f(x) = 2x^2 - 6x + 4$, $g(x) = 4\cos\left(\frac{1}{4}\pi x\right)$



$$A = \int_0^2 [g(x) - f(x)]\,dx$$
$$= \left[4\cdot\frac{4}{\pi}\sin\left(\frac{1}{4}\pi x\right) - \left(\frac{2}{3}x^3 - 3x^2 + 4x\right)\right]_0^2$$
$$= \left(\frac{16}{\pi}\sin\left(\frac{1}{2}\pi\right) - \frac{2}{3}\cdot 8 + 3\cdot 4 - 4\cdot 2\right) - \frac{16}{\pi}\sin 0$$
$$= \frac{16}{\pi} - \frac{4}{3}$$

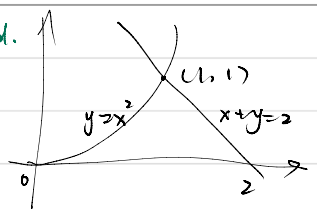
c. $y = 2 - x^2$, $y = -x$



$$2 - x^2 = -x$$
$$x_1 = -1,\ x_2 = 2$$
$$\int_{-1}^2 [2 - x^2 - (-x)]\,dx$$
$$= \left[2x - \frac{1}{3}x^3 + \frac{1}{2}x^2\right]_{-1}^2$$
$$= \left(4 - \frac{8}{3} + 2\right) - \left(-2 + \frac{1}{3} + \frac{1}{2}\right)$$
$$= 6 - \frac{16}{6} + 2 - \frac{5}{6}$$
$$= 8 - \frac{7}{2}$$
$$= \frac{9}{2}$$

top curve changes

d.



① vertical partition

$A = \int_0^1 x^2\,dx + \int_1^2 (2-x-0)\,dx$

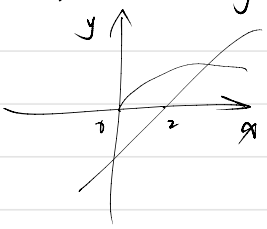
② horizontal partition

↳ $x+y=2$ is always the top curve in terms of y

$A = \int_0^1 [(2-y) - \sqrt{y}]\,dy$

↳ $A = \int_c^d [f(y) - g(y)]\,dy$

e. bound above by $y=\sqrt{x}$, below by $y=x-2$

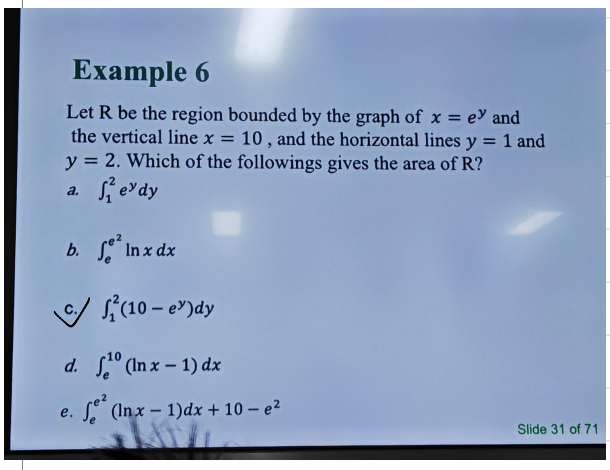


horizontal partition:

$\int_0^2 [2+y-y^2]\,dy$

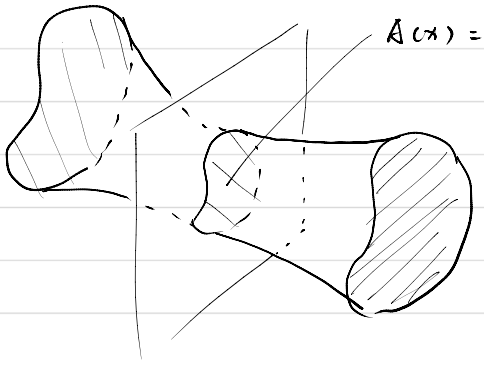
vertical:

$\int_0^2 \sqrt{x}\,dx + \int_2^4 [\sqrt{x} - (x-2)]\,dx$



Find volume using cross-section

cylindrical solid



$A(x)$ = Area of the cross-section

Volume of the kth slab $\approx V_k = A(x_k)\,\Delta x_k$

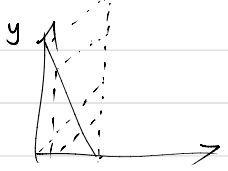
$$V \approx \sum_{k=1}^{n} V_k = \sum_{k=1}^{n} A(x_k)\,\Delta x_k$$

$$V = \lim_{n\to\infty} \sum_{k=1}^{n} A\,x_k\,\Delta x_k = \int_a^b A(x)\,dx$$

square & rectangular

a. base of a solid : bounded by $y = 1 - \frac{x}{3}$, x- and y- axes

each cross-section perpendicular to the x-axis is a square.



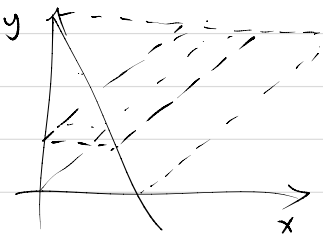
$$A(x) = (s(x))^2$$

$$s(x) = 1 - \frac{x}{3}$$

$$A(x) = \left(1 - \frac{x}{3}\right)^2$$

$$\int_0^3 A(x)\,dx = \int_0^3 \left(1 - \frac{x}{3}\right)^2 dx = \left[-\left(1 - \frac{x}{3}\right)^3\right]_0^3 = 1$$

b. base : bounded by $y = 2 - \frac{2x}{3}$, x- and y- axes



$$x = \frac{3}{2}(2-y)$$

$$A(x) = 5\,(s(y))^2$$

$$= 5\left(\frac{3}{2}(2-y)\right)^2$$

$$\int_0^2 A(y)\,dy = \int_0^2 5\left(\frac{3}{2}(2-y)\right)^2 dy$$

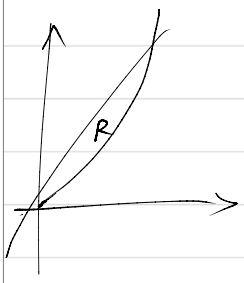
$$= \frac{45}{4}\int_0^2 (2-y)^2\,dy$$

$$= \frac{45}{4}\left[-\frac{1}{3}(2-y)^3\right]_0^2$$

$$= -\frac{15}{4}\left[(2-y)^3\right]_0^2$$

$$= -30$$

c. base : bounded by $y = 4x$ and $y = x^3$, cross-section : perpendicular to y-axis square

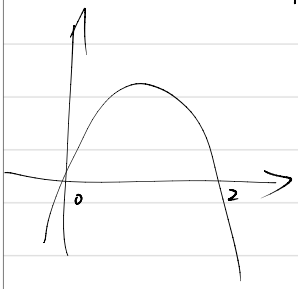


$A(x) = \left(\sqrt[3]{y} - \frac{y}{4}\right)^2$

$\int_0^8 A(y)\,dy$

d. base : bounded by $y = 2x - x^2$ and the x-axis :

cross-section : perpendicular to x-axis, square



$A(x) = (2x - x^2)^2 = 4x^2 - 4x^3 + x^4$

$\int_0^2 A(x)\,dx = \int_0^2 (4x^2 - 4x^3 + x^4)\,dx$

$= \left[\frac{4}{3}x^3 - x^4 + \frac{1}{5}x^5\right]_0^2$

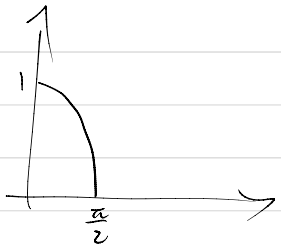
$= \frac{32}{3} - 16 + \frac{32}{5}$

$= \frac{160}{15} - \frac{240}{15} + \frac{96}{15}$

$= \frac{16}{15}$

(calculator) e. base : bounded by $y = \cos x$, x- and y-axes for $0 \le x \le \frac{\pi}{2}$

cross-section : perpendicular to y-axis, rectangular, height = 4 width in xy-plane



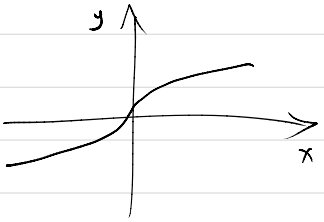
$x = \cos^{-1} y$

$A(y) = 4(\cos^{-1} y)^2$

$\int_0^1 4(\cos^{-1} y)^2\,dy \approx 9.87$

(calculator) f. base : bounded by $y = \tan^{-1} x$, y-axis, $y = 3$, $x = 1$

cross-section : perpendicular to x-axis, square



$s(x) = 3 - \tan^{-1} x$

$A(x) = [3 - \tan^{-1} x]^2$

$V = \int_0^1 (3 - \tan^{-1} x)^2\,dx \approx 6.612$

triangles & semicircles

a. base: bounded by $y = x^2$, $x = 4$, x-axis

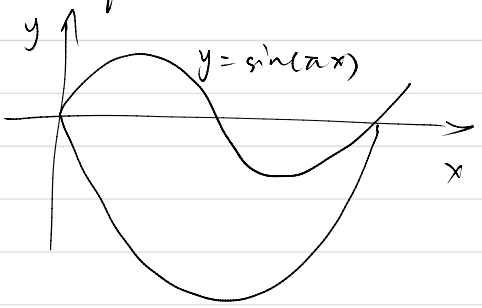
cross-sections: perpendicular to x-axis, equilateral triangles

semicircle: $A(x) = \frac{1}{2}\pi\left(\frac{d}{2}\right)^2$

$d(x) = x^3$

b. base: bounded by $y = x^3 - 4x$, $y = \sin(\pi x)$

depth: $h(x) = 3 - x$



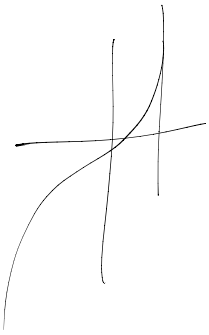
$$\int_0^2 \left[\sin(\pi x) - (x^3 - 4x)\right](3 - x)\,dx$$

c. base: first quadrant, $y = x^2$, x-axis's for $0 \le x \le 1$

cross-section: $\perp$ to x-axis's, a quarter circle,

center on x-axis's, one radius in xy-plane

$$\int_0^1 \frac{1}{4}\pi(x^2)^2\,dx$$



d. R: first quadrant, bounded by $y = x^3$, $x = 2$, x-axis's

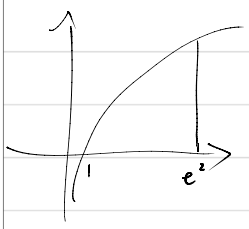
cross-section: perpendicular to x-axis's, equilateral triangle.

$S(x) = x^3$

$\int_0^2 (x^3)^2 \sin 60 \cdot \frac{1}{2}\,dx$

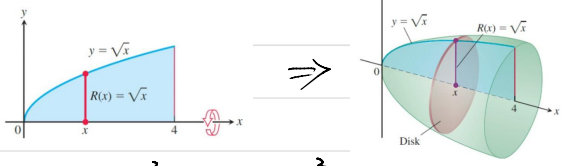
$= \frac{32}{7}\sqrt{3}$

f.



$$\int_1^{e^2} \frac{1}{2}\pi \left(\frac{\ln x}{2}\right)^2 dx$$

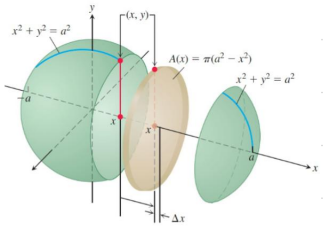
The disk method



$$A(x) = \pi r^2 = \pi (R(x))^2$$

$$V = \int_a^b A(x)\,dx = \int_a^b \pi (R(x))^2\,dx$$

circle rotate about x-axis:



$$R(x) = \sqrt{a^2 - x^2}$$

$$V = \int_{-a}^{a} \pi \left(\sqrt{a^2 - x^2}\right)^2 dx$$

$$= 2\int_0^a \pi \left(\sqrt{a^2 - x^2}\right)^2 dx$$

a.

$$R(x) = 2x$$

$$V = \int_0^3 \pi (2x)^2\,dx$$

$$= 4\pi \int_0^3 x^2\,dx$$

$$= \frac{4}{3}\pi \left[x^3\right]_0^3$$

$$= 36\pi$$

b.

$$R(x) = e^x$$

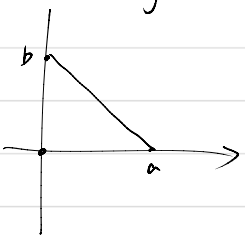
$$V = \int_0^{\ln 2} \pi (e^x)^2\,dx$$

$$= \pi \int_0^{\ln 2} e^{2x}\,dx$$

$$= \frac{1}{2}\pi \left[e^{2x}\right]_0^{\ln 2}$$

$$= \frac{3}{2}\pi$$

c. R: triangular, vertices: (0,0) (a,0) (0,b), revolve about x-axis.



$y = -\frac{b}{a}x + b$

cone: $V = \frac{1}{3}\pi r^2 \cdot h$

$$\int_0^a \pi \left(-\frac{b}{a}x+b\right)^2 dx$$

$$= \pi \int_0^a \left(-\frac{b}{a}x+b\right)^2 dx$$

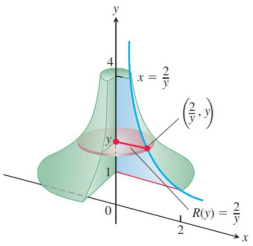
$$= \frac{1}{3}\pi \cdot \left(-\frac{a}{b}\right)\left[\left(-\frac{b}{a}x+b\right)^3\right]_0^a$$

$$= -\frac{1}{3}\pi\frac{a}{b}(-b^3)$$

$$= \frac{1}{3}\pi a b^2$$

Revolving around y-axis's

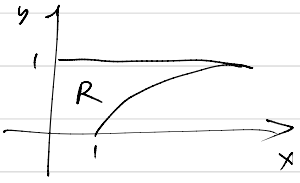
a. between y-axis, $x = \frac{2}{y}$, $1 \le y \le 4$, about y-axis



b. R: bounded by x- and y-axis, $y=1$, $y = \ln x$

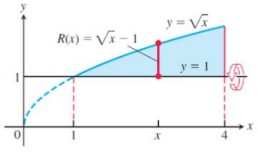
$x = e^y$

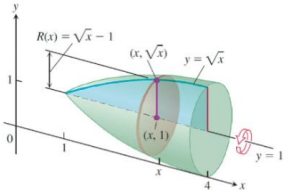
$V = \int_0^1 \pi (e^y)^2 dy$



Revolve around other axes

a. R: $y = \sqrt{x}$, $y=1$, $y=4$, about line $y=1$



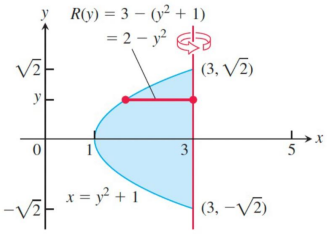


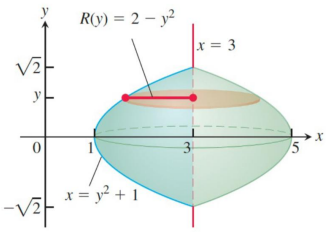
$V = \int_1^4 \pi (\sqrt{x} - 1)^2 dx$ => expand

b. $R: x = y^2 + 1$, $x = 3$, about $x = 3$

$$\int_{-\sqrt{2}}^{\sqrt{2}} \pi (2 - y^2)^2 \, dy$$

$$= 2 \int_{0}^{\sqrt{2}} \pi (2 - y^2)^2 \, dy$$



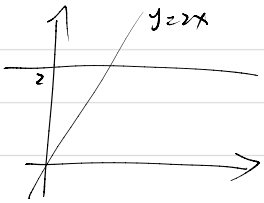


c.

$$x = \tan^{-1} y$$

$$\int_{0}^{\tan 1} \pi (1 - \tan^{-1} y)^2 \, dy$$

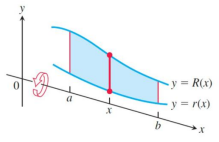
d.

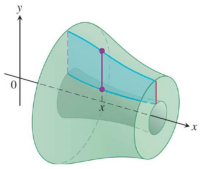


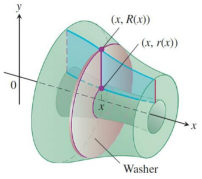
$$\int_{0}^{1} \pi (2 - 2x)^2 \, dx$$

Volume with washer method

$$V = \int_a^b A(x) \, dx = \int_a^b \pi \left(R^2(x) - r^2(x) \right) dx = \pi \int_a^b R^2 - r^2 \, dx$$







a. R bounded by $y = \frac{2}{\pi} x$, $y = \sin x$, about $x = -3$

$R(y) = \frac{\pi}{2} y$ $\quad r(y) = \sin^{-1} y$

$$V = \pi \int_0^1 \left(\left(\frac{\pi}{2} y + 3 \right)^2 - \left(\sin^{-1} y + 3 \right)^2 \right) dy$$

The arc length of a smooth planar curve

if $y = f(x)$, x on $[a,b]$

$L = \int_a^b \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$

if $x = g(y)$, y on $[c,d]$

$L = \int_c^d \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dy$

in terms of x

a. $y = \frac{4\sqrt{2}}{3}x^{\frac{3}{2}} - 1$, $0 \le x \le 1$

$L = \int_0^1 \sqrt{1+(2\sqrt{2}x)^2}\,dx$

$= \int_0^1 \sqrt{1+8x}\,dx$

$= \frac{2}{3}\cdot\frac{1}{8}\left[(1+8x)^{\frac{3}{2}}\right]_0^1$

$= \frac{1}{12}(27-1)$

$= \frac{13}{6}$

b. $L = \int_0^4 \sqrt{1+16x^6}\,dx$

$y' = 4x^3$

$y = \int 4x^3\,dx = x^4 + c$

substitute $(0,3)$, $c = 3$

in terms of y

c. $f(y) = 4(2-y)^2$, $\frac{3}{2} \le y \le 3$

$\int_{\frac{3}{2}}^3 \sqrt{1+(-8(2-y))^2}\,dx$

$= \int_{\frac{3}{2}}^3 \sqrt{1+64(2-y)^2}\,dx$

calculator + practice

d. $y = x + 2\sin x$, $0 \le x \le 3\pi$

$L = \int_0^{3\pi} \sqrt{1+(1+2\cos x)^2}\,dx$

$= \int_0^{3\pi} \sqrt{2+4\cos x+4\cos^2 x}\,dx$

≈ 17.4

e. R bounded by $f(x) = e^{2x}$, $x = k$ $(k>0)$, x- and y- axes, perimeter of R?